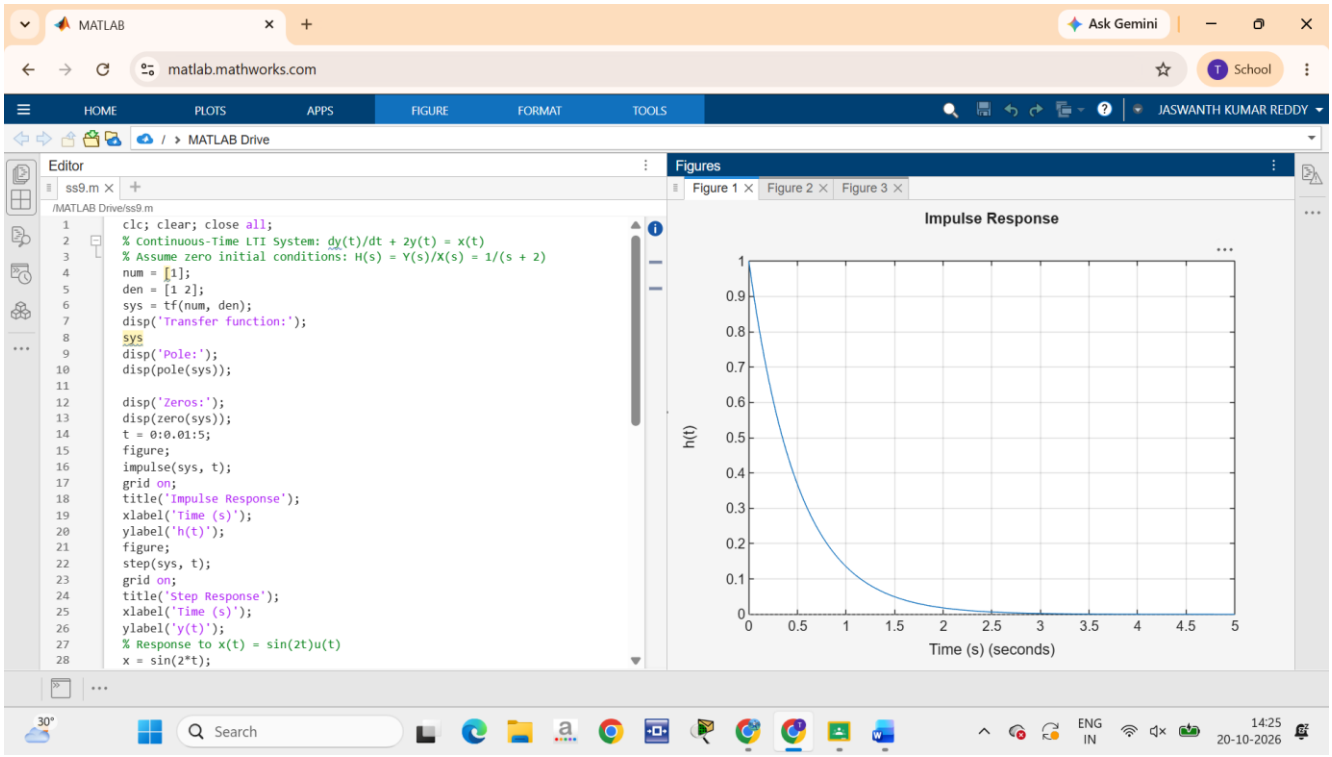
Experiment- 9

Continuous-Time LTI System Modeling Using Differential Equations

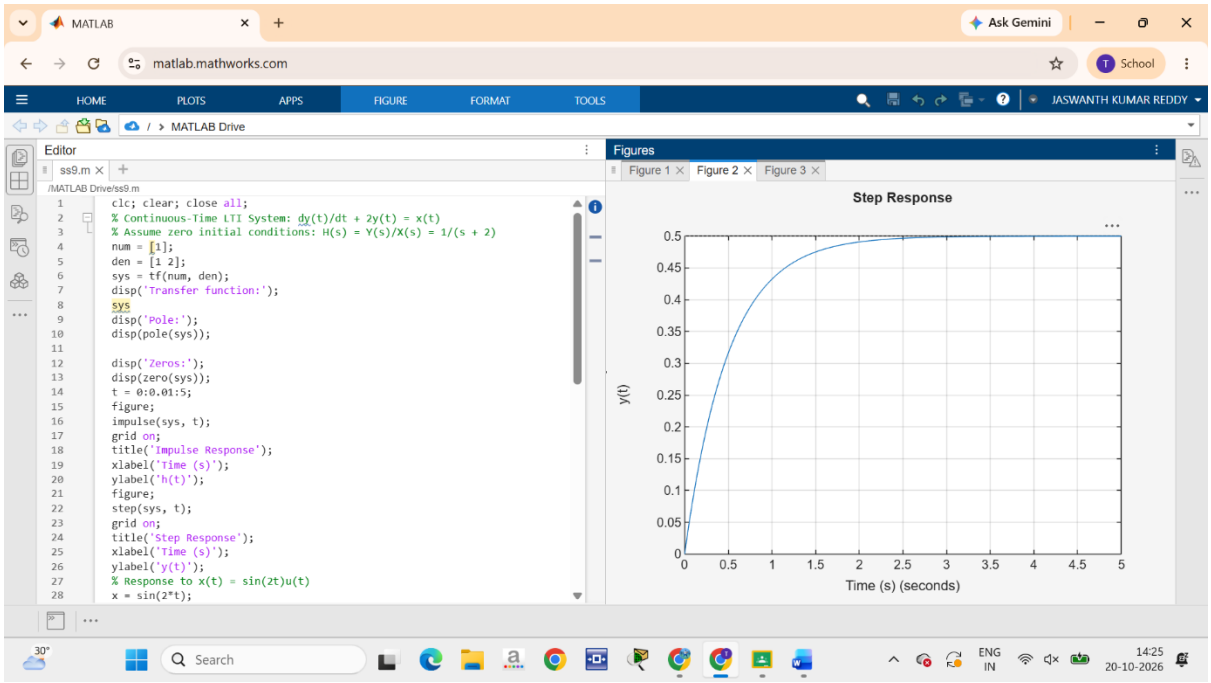
OBJECTIVE:

• To formulate a continuous-time LTI system from a differential equation, obtain its transfer function under zero initial conditions, simulate impulse and step responses, and relate the system poles to its transient behavior and stability.

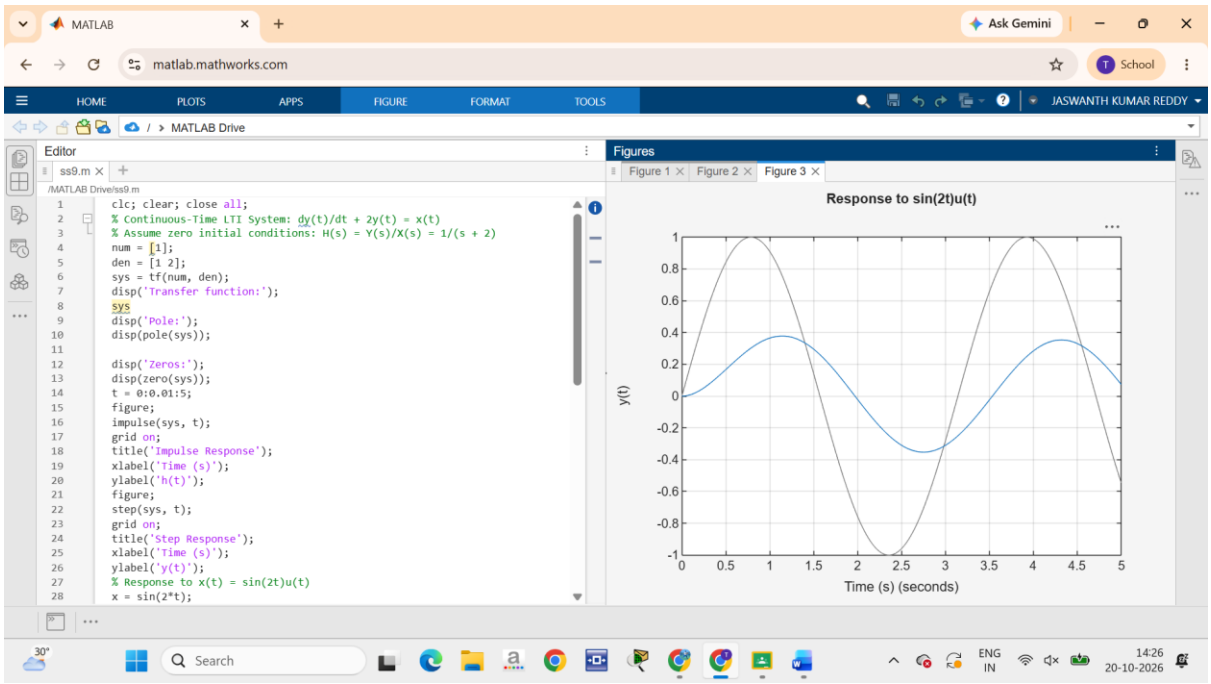
Impulse test:



Step test:



Sinusoidal test:



Result: Continuous-time linear time-invariant (LTI) systems were modeled using differential equations, and their responses were analyzed using MATLAB.